

Circuit emulation

`pb3-replacement.cir` is a system-level ngspice model of the proposed pack. It approximates the TP4056 charger, 1S LiPo, DW01A/FS8205A protection, 2.0 V buck regulator, 200 mA deck load, and output clamp with behavioral sources. It is useful for checking operating points and assumptions. It does not validate the real IC compensation network, stability, thermal behavior, or the unverified AIWA connector pinout.

Run it with ngspice:

```
ngspice -b -o pb3-replacement.log pb3-replacement.cir
grep -E 'vout_|load_current|vbat' pb3-replacement.log
```

Expected nominal results are approximately 1.98 V at the deck output, 198 mA load current, and 3.7 V at the idealized cell. Verify the real circuit with vendor SPICE models and bench measurements before connecting an AIWA deck.